OPPORTUNITY BRIEF · SALES AND PRESALES

Knowledge-graph-based semantic interoperability for digital product passports in electronics manufacturing, enabling automated EU compliance and circular supply chain traceability.

Manufacturing × Product traceability and digital product passport × Knowledge graph and semantic modelling · OS004 v4 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
67 is the world moving	**36** can we play, can we win	**MEDIUM** 1 named in the evidence	**€829m** partial evidence · per year	**NEXT** regulation adopted or propose...	**L3** 16 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

We help electronics manufacturers turn the EU's Digital Product Passport mandate into a competitive advantage by building a knowledge graph that makes product lifecycle data machine-readable and interoperable across your supply chain. This enables automated compliance reporting and circular economy traceability, reducing the burden of fragmented data and manual compliance work.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	**€1.2bn** €147m – €7.6bn	**€829m** €105m – €5.4bn	**€1.0m** €132k – €6.8m	partial
Observed: tendered contract value, annualised	**€20.9m** €20.9m – €56.6m	**€20.9m** €20.9m – €56.6m	**€26k** €26k – €71k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Knowledge graph and semantic modelling	8.7 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_ain2 · E_AI_TTM	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Artificial intelligence by NACE Rev. 2 activity series E_AI_TTM is used as a proxy for Knowledge graph and semantic modelling: Text and language analysis as the nearest series. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

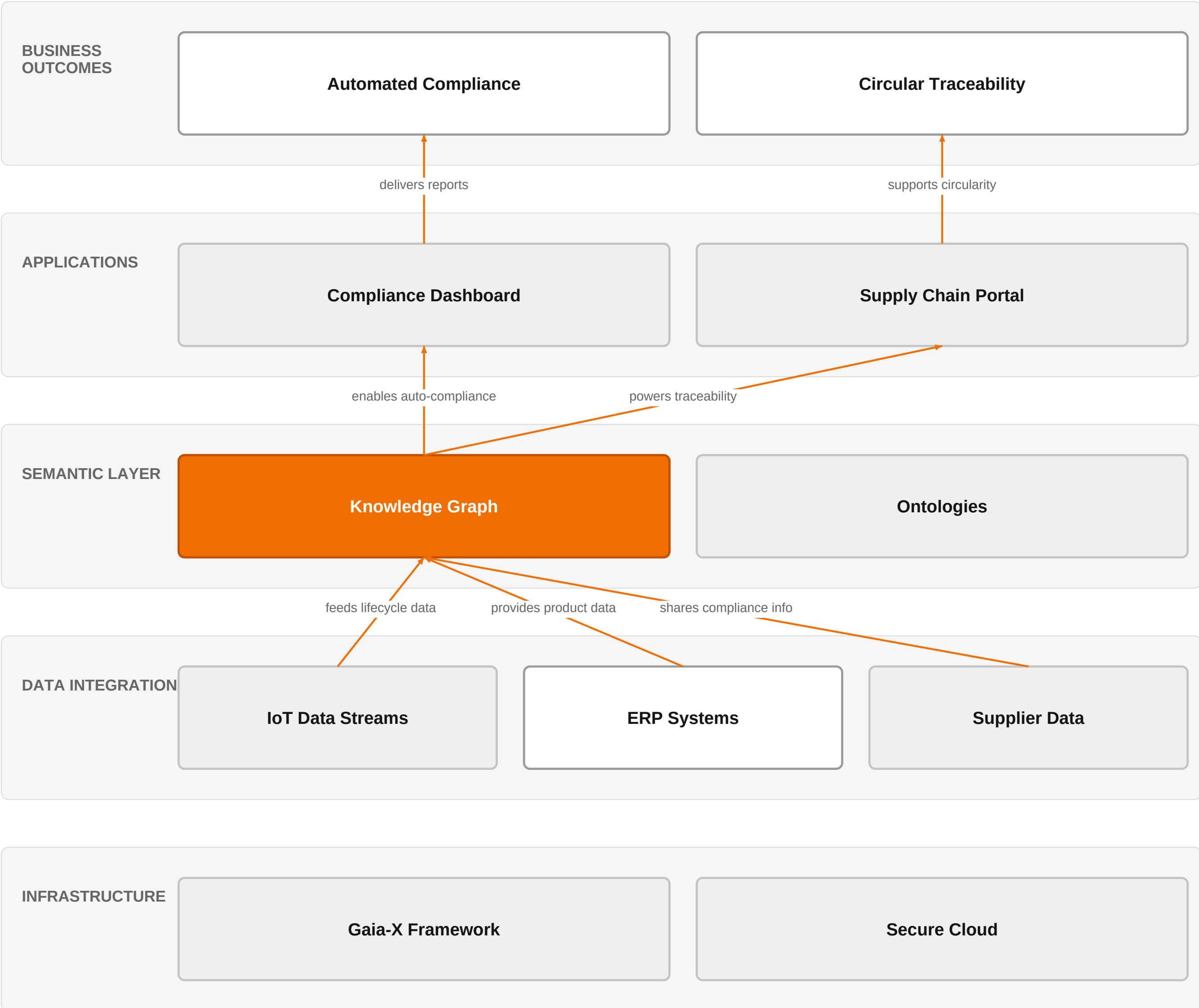
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Knowledge Graph for DPP Compliance



Orange asset · Partner · Customer-owned · Third party / to be sourced

This diagram shows how a knowledge graph layer integrates diverse data sources to enable automated compliance and circular traceability for digital product passports.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU's Ecodesign for Sustainable Products Regulation (ESPR) mandates Digital Product Passports for electronics and appliances by 2027, creating an urgent need for machine-readable product lifecycle data. Semantic Web vocabularies and Semantic Integration Patterns enable interoperability across supply chains, but the landscape of ontologies remains scattered. Gaia-X frameworks provide a proven basis for transparent, interoperable data exchange with auto-compliance tools.

`Sources: SIG-BA196FF05244 Springer Science and B 2026-02-03; SIG-9F888B9F6305 openalex.org 2026-01-01; SIG-63476A345BB2 cordis.europa.eu 2026-06-01`

Who buys, and why now

The chief sustainability officer or head of compliance signs, driven by the ESPR deadline. The pain is felt by supply chain and IT teams who must assemble lifecycle data from fragmented suppliers, often manually, to prove compliance. The trigger is the approaching 2027 mandate and the realization that current systems cannot handle the data volume or interoperability requirements.

Why it is hot now — every claim bound to a dated source

- Semantic Integration Patterns facilitate interoperability in Digital Product Passports, enabling traceability and lifecycle data sharing across supply networks. [SIG-9F888B9F6305]
- Digital Product Passports can support circular and collaborative ecosystems in fashion and cosmetics by enhancing transparency, traceability, and knowledge exchange. [SIG-6B43CC88BC8E]
- Semantic Integration Patterns (SIPs) provide reusable communication patterns for structured data sharing in Digital Product Passports. [SIG-9F888B9F6305]
- Semantic Integration Patterns (SIPs) facilitate semantic interoperability in DPPs for traceability and lifecycle data sharing. [SIG-9F888B9F6305]
- Semantic Integration Patterns facilitate semantic interoperability in DPPs for traceability and compliance across supply networks. [SIG-9F888B9F6305]
- DPPs serve as digital representations of physical products, enabling traceability and lifecycle data sharing. [SIG-9F888B9F6305]
- Gaia-X frameworks ensure transparency and interoperability in data exchange with auto-compliance tools along the value chain. [SIG-63476A345BB2]
- Semantic Web vocabularies can enable interoperability across supply chains for DPP, yet the landscape of ontologies covering DPP remains scattered and not systematically compared. [SIG-BA196FF05244]

What Orange would deliver, and why it can win

What Orange would deliver

We assemble a knowledge graph and semantic modelling solution, leveraging Orange Group researchers and IoT experts to integrate sensor and lifecycle data. We use ISO 27001 and TISAX certifications to assure security and automotive-grade compliance. The engagement starts with a data landscape assessment, then builds a semantic model using existing ontologies, and finally deploys a Gaia-X-aligned data exchange with auto-compliance tools.

Why Orange can win

Orange brings a unique combination of IoT expertise, research capability, and enterprise-grade certifications (ISO 27001, TISAX). Our experience with large manufacturers like Colgate-Palmolive, Mondelez, and Saint-Gobain Glass demonstrates our ability to handle complex supply chain data. However, our knowledge graph offering is still developing; we must be honest that our semantic modelling capability is not yet as mature as specialist competitors.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Knowledge graph and semantic modelling	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical for this use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 4.0; 1 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Databricks offers a data and AI platform that can handle large-scale data integration, but lacks deep manufacturing domain expertise. IBM and Palantir sell knowledge graph and semantic modelling solutions with strong enterprise track records. Accenture and Capgemini bring systems integration muscle and manufacturing focus. Deutsche Telekom and Vodafone offer telecom infrastructure but less semantic depth. Bosch has industrial OT expertise but is a potential partner or competitor.

`Sources: SIG-BC952EFA76FC Databricks 2026-07-27`

Competitor	Category	Position here	Basis	Evidence
Databricks	Data and AI platform	named in this topic's evidence	evidenced	SIG-BC952EFA76FC
IBM	Global systems integrator	sells Knowledge graph and semantic modelling	structural	—
Palantir	Data and AI platform	sells Knowledge graph and semantic modelling; competes in OX: Smart Industries	structural	—
Accenture	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Manufacturing; competes in OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	active in Manufacturing; competes in OX: Smart Industries	structural	—
Bosch	Industrial / OT specialist	active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

1. Which of your product lines are in scope for the EU Digital Product Passport regulation by 2027, and what is your current compliance readiness?
2. How do you currently collect and share product lifecycle data across your supply chain—is it manual, fragmented, or already digitized?
3. Have you already evaluated any semantic web vocabularies or ontologies for DPPs, and if so, which ones and what gaps did you find?
4. What is your current approach to exchanging data with suppliers and customers—do you use any interoperability frameworks like Gaia-X?
5. Who in your organization owns the DPP compliance project, and what is their budget and timeline?
6. Do you have existing IoT or sensor data from your products that you intend to feed into the DPP, and how is it currently managed?

Objections you will meet

They will say	You can answer
We already have a data platform from Databricks or similar; why do we need a knowledge graph?	That's a fair point—data platforms are great for storing and processing data, but they don't inherently provide the semantic interoperability that DPPs require. A knowledge graph adds a layer of meaning that enables automated compliance and cross-supply-chain data sharing, which a traditional platform alone may not deliver.
The regulation is still evolving; we're not sure about the final requirements.	You're right, the technical specifications are still being finalized. However, the ESPR mandate is clear, and starting with a flexible semantic model based on existing vocabularies positions you to adapt quickly. Our approach is designed to be future-proof, so you won't have to start from scratch.

We don't have the internal expertise to manage a knowledge graph project.	That's exactly why Orange is a good partner. We bring not only the technology but also the research and IoT expertise to guide you through the process. We can also leverage our experience from similar projects with large manufacturers to accelerate your implementation.

Risks and unknowns

The main risk is that the customer may not yet feel the urgency if their products are not in the first wave of ESPR. There is also uncertainty about the final regulatory technical specifications for DPPs, which could change the required data model. Our own knowledge graph capability is still maturing, and we may need to partner for specialized ontology design.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a digital product passport project in electronics manufacturing, leveraging our knowledge graph and semantic modelling technology, and referencing our ISO 27001 certification and experience with clients like Colgate-Palmolive and Saint-Gobain Glass to demonstrate credibility.
Sales	In a meeting with a manufacturing client's head of sustainability or digital transformation, say: 'We have deep expertise in IoT and semantic modelling that can help you prepare for the upcoming EU digital product passport requirements, ensuring automated compliance and traceability across your supply chain.'
Strategist / Innovator	Commission a deep-dive brief on knowledge-graph-based semantic interoperability for digital product passports, focusing on the electronics manufacturing vertical and mapping the scattered ontology landscape, to decide whether to invest in building a reusable semantic layer.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-63476A345BB2	2026-06-01	cordis.europa.eu	1	Fostering Resilient and Auto-Compliant Manufacturing Value Chains through Advanced Digital Prod...
SIG-6B43CC88BC8E	2026-03-14	openalex.org	1	Integrating knowledge management and digital product passports to foster sustainable and collab...
SIG-F47C31AB60CB	2026-02-14	MDPI AG	1	Dynamic Traceability and Reliability Management: Integrating the Digital Product Passport with ...
SIG-BA196FF05244	2026-02-03	Springer Science and Busi...	1	Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...
SIG-7B174A2FD055	2026-01-29	MDPI AG	1	Verisav Vocabularies: RDF/OWL Ontologies for Digital Product Passport and After-Sales Service A...
SIG-4B5C641F1D4C	2026-01-01	Elsevier BV	1	Digital Product Passport and Internet of Things Integration in Supply Chain Management
SIG-4FE21FD2ED06	2026-01-01	openalex.org	1	On the Utilization of Digital Product Passports for Intelligent Metal Sorting and Recycling
SIG-53ADAC7DA5A5	2026-01-01	RSC Sustainability	1	Supply chain optimisation using physics-informed machine learning for digital product passport
SIG-9F888B9F6305	2026-01-01	openalex.org	1	Semantic Data Integration for Digital Product Passports
SIG-CC28A63D0BB7	2026-01-01	openalex.org	1	From Compliance to Circularity: Stakeholder Perspectives on Adopting Digital Product Passports ...
SIG-40DC35D6534F	2025-10-14	openalex.org	1	Predictive Generation of Digital Product Passport as Decision Support in Circular Networks
SIG-F0C3FBE1B02D	2026-08-18	The Chartered Institute o...	2	Government looks for businesses to input on digital product passports - The Chartered Institute...
SIG-BC952EFA76FC	2026-07-27	Databricks	2	The EU Digital Product Passport: a traceability deadline - Databricks
SIG-BE9AC623C90C	2026-07-02	Cambridge University Pres...	2	Towards a reference architecture for digital product passports in engineering - Cambridge Unive...
SIG-D4A38DBA082C	2026-06-30	Engineering News	2	Prepare for the future of trade: Understanding Digital Product Passports - Engineering News
SIG-A3AAC433E98C	2026-05-07	HKTDC Research	2	Proposed EU Rules Clarify Operation of Digital Product Passport Registry - HKTDC Research

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:13+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:16+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:40:26+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.